

scripts/autonomous-qa/lib-remote.sh — User Guide

Last verified: 2026-07-26 (LVA-018 script-docs backfill cycle)
Inheritance: HelixConstitution §11.4.18 (script docs); Lava §6.H (no secret echo), §6.J (failures surface), §6.R (config from .env), §6.U (no sudo) **Classification:** project-specific

Overview

Sourceable library implementing **Approach B (run-on-thinker)**: co-locate the ENTIRE autonomous-QA heavy stack on thinker.local so build (gradle) + containerized KVM emulator (podman) + adb + connectedAndroidTest + backend all share thinker’s loopback. This is the only topology where gradle’s adb reaches the emulator and the on-emulator client reaches the backend without WAN/adb tunnels — the Containers submodule’s pkg/emulator/containerized.go drives LOCAL podman only (no SSH path), so the emulator **MUST** be local to whichever host runs the matrix; that host is thinker (/dev/kvm + 16 cores).

Thin glue over ssh + rsync, keyed on the same identity that authenticates today (~/.ssh/id_ed25519, confirmed via ssh -G).

Functions

Function	Purpose
remote_preflight	Prove reachability + key auth + podman + rsync + /dev/kvm before any heavy work is dispatched; non-zero (with explanation) if thinker is unfit
remote_sync_repo	rsync the repo → thinker (excludes build/.git/recordings, INCLUDES the gitignored .env so the Go backend’s LAVA_AUTH_* + tracker creds reach thinker); --delete keeps the remote tree a faithful mirror

Function	Purpose
<code>remote_run "<cmd>"</code>	Run a command on thinker with <code>cwd</code> = the synced repo; stdout/stderr stream straight back
<code>remote_fetch <remote-rel-path> <local-dest></code>	Pull curated evidence back

Configuration (from `.env`, gitignored — config, not secrets per §6.R)

Variable	Default	Meaning
<code>LAVA_API_GO_REMOTE_HOST</code>	<code>thinker.local</code>	QA host
<code>LAVA_REMOTE_HOST_USER</code>	<code>milosvasic</code>	SSH user
<code>LAVA_REMOTE_HOST_KEY</code>	<code>~/.ssh/id_ed25519</code>	SSH identity (path only; contents never read/printed)
<code>LAVA_THINKER_QA_DIR</code>	<code>/home/<user>/lava-qa</code>	Remote repo location

SSH behavior

Shared options: `BatchMode=yes` (key-only, never prompt), bounded connect (10 s), `StrictHostKeyChecking=accept-new`, `keepalives`, and a persistent ControlMaster (`ControlPersist=300`, sockets under `$ {TMPDIR:-/tmp}/lava-qa-ssh-ctrl`) so the many `remote_run` calls in a matrix reuse one TCP/SSH session.

Security posture

- §6.H: no secret is ever echoed. `.env` travels over the encrypted SSH channel, never logged. The key path is config; its contents are never read.
- §6.U: no `sudo/su` anywhere on either side.
- §6.J: `set -euo pipefail`; failures surface, never swallowed.

Usage

```
source scripts/autonomous-qa/lib-remote.sh
remote_preflight
remote_sync_repo
```

```
remote_run "scripts/autonomous-qa/run-matrix.sh --backend goapi --  
           subsets rutracker --queries 1080p"  
remote_fetch ".lava-ci-evidence/autonomous-qa/$(date +%F)/goapi/  
            summary.md" /tmp/summary.md
```

Companion files

- scripts/autonomous-qa/run-matrix.sh — the matrix typically dispatched via remote_run
- scripts/distribute-api-remote.sh (+ its doc) — the sibling thinker remote-distribute path